WORKED SOLUTION

STEP II 1993, Question 9

Pure • Arguments of Complex Numbers

STEP II 1993, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q9

The key idea

With the convention $0 \leq \arg z < 2\pi$, "positive imaginary part" means "argument strictly between 0 and π". Adding complex numbers is adding vectors, and a sum of vectors that all point into the same open half-plane stays inside it; rotating so that z_1 or z_3 lies along the real axis turns the inequalities into exactly this fact. For the last part, the argument of a product is the sum of the arguments (up to a multiple of 2π), and the inequalities just proved trap the difference strictly between 0 and 2π.

Solution

Positive imaginary part

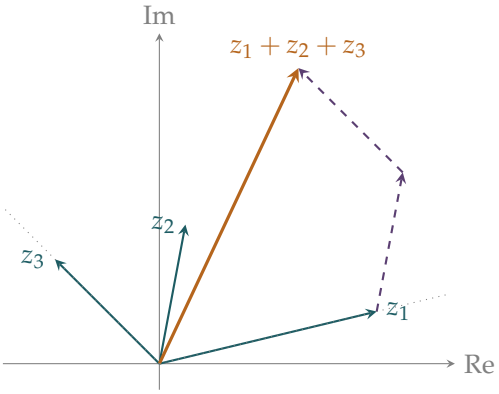
Write $z = r(\cos\theta + \mathrm{i}\sin\theta)$ with $r > 0$ and $0 \leq \theta < 2\pi$. Then $\operatorname{Im} z = r\sin\theta > 0$ exactly when $\sin\theta > 0$, and on $[0, 2\pi)$ this happens exactly when

$$0 < \arg z < \pi.$$

So z lies in the upper half-plane, and its argument is never 0 or π (those would make z real).

The diagram

Write $\theta_k = \arg z_k$, so $0 < \theta_1 < \theta_2 < \theta_3 < \pi$. Draw z_1, z_2, z_3 as arrows from the origin; then place them head to tail to reach $z_1 + z_2 + z_3$.



Each step of the path turns anticlockwise relative to the previous one (because $\theta_1 < \theta_2 < \theta_3$), and every step points into the upper half-plane. The path starts along the ray at angle θ_1 and then bends to the left of it, so its endpoint lies strictly anticlockwise of that ray; looking back from the other end, the path finishes with a step at angle θ_3, and everything before it lies clockwise of the ray at angle θ_3. So the sum lies in the wedge between the rays at angles θ_1 and θ_3.

To make this precise, multiply by $\mathrm{e}^{-\mathrm{i}\theta_1}$, which rotates everything clockwise by θ_1:

$$\mathrm{e}^{-\mathrm{i}\theta_1}(z_1 + z_2 + z_3) = r_1 + r_2\mathrm{e}^{\mathrm{i}(\theta_2-\theta_1)} + r_3\mathrm{e}^{\mathrm{i}(\theta_3-\theta_1)},$$

where $r_k = |z_k| > 0$. The first term is real and positive, and the other two have angles in $(0, \pi)$, so positive imaginary part. Hence the right-hand side has positive imaginary part, and its argument α satisfies $0 < \alpha < \pi$. Therefore $z_1 + z_2 + z_3$ has an argument $\theta_1 + \alpha$, and since $\theta_1 < \theta_1 + \alpha < \theta_1 + \pi < 2\pi$ this is the argument in the range $[0, 2\pi)$. So $\arg(z_1 + z_2 + z_3) > \theta_1$.

Similarly, multiplying by $\mathrm{e}^{-\mathrm{i}\theta_3}$ gives $r_1\mathrm{e}^{\mathrm{i}(\theta_1-\theta_3)} + r_2\mathrm{e}^{\mathrm{i}(\theta_2-\theta_3)} + r_3$, where the first two angles lie in $(-\pi, 0)$, so this has negative imaginary part and can be written $\rho\,\mathrm{e}^{-\mathrm{i}\beta}$ with $0 < \beta < \pi$. Then $z_1 + z_2 + z_3 = \rho\,\mathrm{e}^{\mathrm{i}(\theta_3-\beta)}$, where $-\pi < \theta_3 - \beta < \theta_3 < \pi$. The sum has positive imaginary part, so its argument lies in $(0, \pi)$ and differs from $\theta_3 - \beta$ by a multiple of 2π; as both numbers lie in $(-\pi, \pi)$, an interval of length 2π, they are equal. So $\arg(z_1 + z_2 + z_3) = \theta_3 - \beta < \theta_3$, and altogether

$$\arg z_1 < \arg(z_1 + z_2 + z_3) < \arg z_3.$$

The product and the sum

Let $\varphi = \arg(z_1 + z_2 + z_3)$, so $\theta_1 < \varphi < \theta_3$ by the previous part. Multiplying complex numbers adds their arguments, so

$$z_1z_2z_3 = r_1r_2r_3\,\mathrm{e}^{\mathrm{i}(\theta_1+\theta_2+\theta_3)},$$

and $\arg(z_1z_2z_3)$ is the number in $[0, 2\pi)$ that differs from $\theta_1 + \theta_2 + \theta_3$ by a multiple of 2π.

Ask yourself The sum $\theta_1 + \theta_2 + \theta_3$ can be anywhere in $(0, 3\pi)$, so we don't know which multiple of 2π to subtract. How can we avoid needing to know?

We don't need to know: we only need that if $\arg(z_1 z_2 z_3) = \varphi$ then

$$D = \theta_1 + \theta_2 + \theta_3 - \varphi$$

is a multiple of 2π. Now bound D using $\theta_1 < \varphi < \theta_3$:

$$D > \theta_1 + \theta_2 + \theta_3 - \theta_3 = \theta_1 + \theta_2 > 0,$$
$$D < \theta_1 + \theta_2 + \theta_3 - \theta_1 = \theta_2 + \theta_3 < 2\pi,$$

using $\theta_k > 0$ in the first line and $\theta_k < \pi$ in the second. So $0 < D < 2\pi$, and no multiple of 2π lies strictly between 0 and 2π. Hence $\arg(z_1 z_2 z_3)$ is never equal to $\arg(z_1 + z_2 + z_3)$.

Checks

For the diagram's example, $z_1 = 2.5 + 0.6\mathrm{i}$, $z_2 = 0.3 + 1.6\mathrm{i}$, $z_3 = -1.2 + 1.2\mathrm{i}$, the arguments are about 0.236, 1.385 and 2.356, and $\arg(z_1 + z_2 + z_3) = \arg(1.6 + 3.4\mathrm{i}) \approx 1.131$, which lies between the first and last. In a test of 200 000 random triples (moduli between e^{-4} and e^{4}), both inequalities and $0 < D < 2\pi$ held every time, and the product and sum never pointed in the same direction.

To see why the upper half-plane matters, try $z_1 = z_2 = z_3 = \mathrm{e}^{2\pi\mathrm{i}/3}$ with the ordering relaxed to equality: the product is 1 and the sum points along $\mathrm{e}^{2\pi\mathrm{i}/3}$, consistent with $D = 4\pi/3$. Letting all three angles approach π pushes D towards 2π, so the bound $\theta_k < \pi$ is doing real work.

Common slips

- Writing $\arg(z_1 z_2 z_3) = \theta_1 + \theta_2 + \theta_3$ outright: this is false whenever the sum is 2π or more. Only equality modulo 2π is available.
- Arguing that $\arg(z_1 + z_2 + z_3)$ is some kind of average of the θ_k. It isn't (the moduli matter); what is true is only that it lies strictly between θ_1 and θ_3.

Beyond STEP

The first fact is the statement that an open half-plane (indeed any wedge of angle less than π) is a *convex cone*: positive combinations of vectors in it stay in it. Convex cones are central in linear programming and optimisation, and the multivaluedness of arg that the last part works around leads to branch cuts and the complex logarithm in a first course on complex analysis. Needham's *Visual Complex Analysis* is a beautiful geometric treatment of both ideas.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.